LAB: Molecular Docking and Drug Design

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Introduction

In this lab, you will learn how to use a method called molecular docking. A docking program is used in order to predict the conformation and position of a ligand relative to a protein as well as the binding affinity. Usually the ligand is treated as flexible and the protein as rigid. The docking program generates poses (conformation and position) of the ligand in the active site of a protein structure and uses a scoring function to score how good the poses are. Scoring functions usually predict affinity (ΔG, K_d, IC_{50} ...).

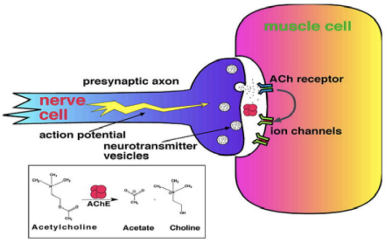
This lab is divided into two sessions. During the first session, we will dock five small molecules to acetylcholinesterase. The small molecules are Tacrine, Donepezil, Galantamine, the natural product Huperzine A, and Ibuprofen. All except for Ibuprofen are acetylcholinesterase binders. During the second session, your job is to design a new drug of your own and to dock it to the protein.

Work in pairs during the two lab sessions and hand in a complete written report before the exam.

Background

Acetylcholine and acetylcholinesterase inhibition

The parasympathetic nervous system has a critical role in our bodies. It acts on smooth muscles, cardiac muscle and glands and takes part in the control of heart rate, digestion, blood pressure and glandular secretion. The system is turned on when the body is at rest.



The parasympathetic nervous system uses acetylcholine as its major neurotransmitter. During transmission of nerve impulses, an action potential moves through the nerve cell to the nerve ends and, by the help of Ca^{2+} ions, stimulates the release of acetylcholine. Acetylcholine diffuses through the synaptic cleft and binds to acetylcholine receptors on the postsynaptic membrane. Binding to these receptors triggers the opening of ion channels and a new action potential is formed. Acetylcholine is decomposed into choline and acetate by an enzyme called acetylcholinesterase, which is present in the synaptic cleft.



Inhibition of acetylcholinesterase disrupts the parasympathetic nervous system with severe implications for the body. The most feared snake in Africa, the Black Mamba, has an acetylcholinesterase inhibiting peptide called Fasciculin as part of its venom. A bite leads to dizziness, breathing difficulty, erratic heartbeat and a certain death within a couple of hours.

Insecticides and chemical weapons like VX gas, Soman and Sarin are also acetylcholinesterase inhibitors. They bind irreversibly to the enzyme. The lethal gas Sarin was used by Iraq in the Kurd city Halabja 1988, where about 7000 people were killed. It has also been used by terrorists, for example by the Aum sect in the Tokyo subways in 1995.

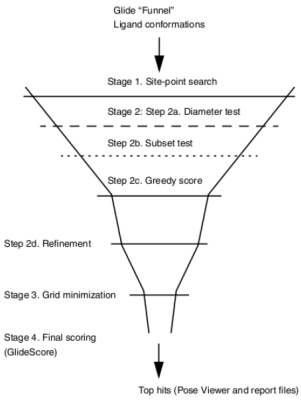
Alzheimer's disease

About 100,000 people in Sweden live with Alzheimer's disease. The disease is incurable, but can be treated with acetylcholinesterase inhibitors. The influence of the disease on the patient and relatives is gigantic and new drugs are needed. During the disease, nerve cells are destroyed and the peptides β-Amyloid form plaque in the cortex. Alzheimer's patients show low concentrations of acetylcholine due to the death of cholinergic neurons. By inhibiting acetylcholinesterase, the concentration increase, which leads to improvements of the patients cognitive functions. The first acetylcholinesterase inhibitor on the market used for Alzheimer was Tacrine (Cognex) in 1993. Tacrine was unregistered in Sweden in 2000 as a result of side effects on the liver at therapeutic doses. Today there are three acetylcholinesterase inhibitors used for treatment in Sweden: Donepezil (Aricept), Galantamine (Reminyl), and Rivastigmine (Exelon).

Docking with Glide

Schrödinger suite has a docking application called Glide (**G**rid based **LI**gand **D**ocking with **E**nergetics), which searches for favourable interactions between the ligand and protein to generate ligand poses. A pose refers to the combination of position and orientation of the ligand relative to the protein, along with its conformation. The generated poses are passed through a series of hierarchical filters that evaluate the ligand's interaction with the protein, to reduce the search problem.

The initial filters perform an exhaustive search to test the spatial fit of the ligand into the defined active site, using a grid-based method. For example, orientations that give too many steric clashes are skipped. These first steps rapidly eliminate unsuitable ligand conformations and only a small number of the best refined poses (typically 100-400) are passed to the final stages in the hierarchy, which involve energy minimisation using a grid approximation to the OPLS-AA nonbonded ligand-receptor interaction energy. Finally, the minimised poses are re-scored using the GlideScore scoring function. GlideScore uses a combination of force field potentials (LJ and Coulomb) and empirical terms like lipophilic interactions, hydrogen-bonding, and terms that penalise steric clashes.



The choice of best-docked structure for each ligand is made using a model energy score (Emodel) that combines the energy grid score, the binding affinity predicted by GlideScore, and (for flexible docking) the internal strain energy for the model potential used to direct the conformational-search algorithm.

Glide has three different modes: HTVS (high throughput virtual screening), SP (single precision) and XP (extra precision). HTVS is the fastest and most uncertain mode while XP is the slowest but also the most powerful method. XP produces more precise poses by using a custom scoring function designed to identify ligand poses that would be expected to have unfavourable energies. This means that the scoring function (GlideScore XP) includes additional terms over the SP scoring function.

Day 1. Docking

Log in to your computer. Copy the glide-folder, which contains the structures needed for the lab, and change to this directory by typing

```
cp -r /home/1MB412/glide .

cd glide
```

Then open Maestro.

Import the structure AchE_Done.pdb from the pdb folder. This is the x-ray crystal structure of the complex between acetylcholinesterase and the inhibitor Donepezil. Now have a look at the complex. Show the ligand in balls and sticks and display only residues within 5 Å of the ligand.

Question 1: *What kind of atoms are the small red dots that are close to the ligand, but not part of the protein?*

We are going to prepare the structure for docking by using the Protein Preparation Wizard tool. Click on Protein Preparation Wizard.

Make sure that the following settings are marked: Assign bond orders, Add hydrogens, Delete waters beyond 5 Å (leave the ones that are already marked). Click on Preprocess. A dialog box will open if any problems have been found with the structure after preprocessing, such as incorrect atom types, missing atoms, overlapping atoms, etc..

Question 2: *Which atoms are missing in the structure and where are they located in the protein?*

In the dialog box, it is possible to add any missing side chains, however, this takes a bit of time. Here, we will ignore missing atoms (motivated by their location) and overlapping atoms. Click OK in the dialog box and go to the refine tab in the Protein Preparation Wizard. Under H-bond assignment, make sure that the Sample water orientations setting is marked. Click on Optimize. Wait until you see that the job is finished in the Jobs monitor window (top right).

Close the Protein Preparation Wizard. Make sure that only the last structure is selected and included in the workspace ("In" marked).

Question 3: *What kind of interactions can you identify between the ligand and the protein?*

For studying some of the types of interactions present, you can hover over the Interactions toggle on the Workspace Configuration Toolbar until the "..." icon

appears above. Click on it to open the Interactions pane, where you can choose to display different types of interactions in the Workspace (shown as dashed lines of different colours).
It is also possible to generate a 2D diagram of the interactions by clicking the Ligand Interaction Diagram panel. Residues are represented as coloured spheres, labeled with the residue name and residue number and coloured according to their properties. The ligand is displayed as a 2D structure. Interactions between the residues and the ligand are drawn as lines, coloured by interaction type. The binding pocket is indicated by a line drawn around the ligand, coloured by the colour of the nearest residue. Select Sync with 3D to automatically update the 2D diagram as you change your camera view in the Workspace.

It is now time for us to generate a grid for the docking.

1. Type "grid generation" in the Tasks menu search field. Click on grid generation and then on Receptor Grid Generation.
2. In the Define receptor section (of the Receptor tab), ensure that Pick to identify ligand and Show markers are selected, and that, in the option menu, Molecule is chosen.
3. In the Workspace, pick an atom in the ligand molecule. Green markers appear on the ligand.
4. In the Van der Waals radius scaling section, ensure that Scaling factor is set to the default value of 1.00 (no scaling.)
5. Click the Site tab. The entire complex is shown with several types of markers: the green ligand molecule markers that appeared when the ligand was identified, and the new markers that appeared when the Site tab was opened (you may have to zoom out to see them):
    - The enclosing box is shown in purple and represents the volume of the protein for which grids will be calculated.
    - The center of the enclosing box is marked by green coordinate axes.
6. In the Site tab, ensure that the Center option selected is Centroid of Workspace ligand.
7. Ensure that the Size option selected is the default, Dock ligands similar in size to the Workspace ligand. Have a look at the structure (zoom out if needed) and see if the box is large enough.
8. Change the Job name to AchE_Done_Grid.
9. Start the job by clicking Run. The job takes about 3 minutes. You can check if it is running in the Jobs menu.

While we wait for the job to finish, import the x-ray crystal structure of the nerve agent sarin in complex with acetylcholinesterase, AchE_Sari.pdb.

Also include the AchE_Done structure in the workspace by Ctrl-click on the In circle in the Workspace Navigator. Display all atoms within 5 Å from the ligand. Use the Structure Hierarchy to find the SGB ligand.

Question 4: *Is there a difference in the acetylcholinesterase structures?*
Question 5: *How do Sarin and Donepezil differ in their binding to acetylcholinesterase?*
Question 6: *What is the difference between irreversible and reversible binding?*

Now we are going to open a number of inhibitors which we are going to shortly dock. Open the Import window and navigate to the inhibitors directory. Select all files (.mae format) in the inhibitors directory and click on Open. Look at the different inhibitors.

Question 7: *What similarities and differences are there between the inhibitors?*

Ensure that the grid-job is finished. Now it is time for the docking.

1. Make sure that the inhibitors (only the inhibitors) are selected in the Project Table.
2. From the Tasks menu, choose Docking > Ligand docking. The Ligand Docking panel opens with the Ligands tab displayed.
3. In the Receptor grid section (top), click the Browse button. A file selector opens.
4. In the directory AchE_Done_Grid, choose AchE_Done_grid.zip.
5. In the Settings tab, change the Precision option to XP (extra precision).
6. In settings, ensure that Ligand sampling flexibility and Sample nitrogen inversions and Sample Ring Conformations are selected, and Penalize nonplanar conformation is chosen from the Amide bonds option menu. (These are the defaults.)
7. In the Ligands tab, set "Use ligands from" to Project Table (selected entries).
8. Ensure that Scaling of van der Waals radii is set to the default values: Scaling factor to 0.80 and Partial charge cutoff to 0.15.
9. Change the Job name to Dock_AchE_Done.
10. Click Run. Ensure that the job is running under the Jobs menu.

The calculations take about 5-10 minutes - time for a break.

Import the structures AchE_Gala.pdb, AchE_Hupe.pdb and AchE_Tacr.pdb from the pdb-folder. These are x-ray crystal structures of acetylcholinesterase complexed with Galantamine, Huperzine A, and Tacrine and can be used as a confirmation that the docking has worked.

Ensure that the docking job is finished. In the Workspace Navigator and Project Table there will be a new group containing the protein structure and final poses of the inhibitors calculated by glide.

You have now docked the ligands to the crystal structure of the acetylcholinesterase, where Donepezil was taken out. Compare the docking solutions to the respective crystal structure, e.g. show the protein you have used for the docking (the empty AchE_Done), the docked Galantamine and the crystal structure of Galantamine bound (AchE_Gala). Do the same for the rest.

Question 8: *Do the docked structures agree with the x-ray crystal structures?*
Question 9: *If not, can you see any reason why the docking did not work?*

Look at the Glide score in the Project table. According to measured IC50-values, the ranking of the binders are HuperzineA > Donepezil > Tacrine > Galantamine (higher affinity > lower affinity). Ibuprofen is not considered to be an inhibitor for acetylcholinesterase. It is one of the more commonly used non steroidal anti-inflammatory drugs (pain killers) and has its effect by binding to the enzyme COX-2. The use of Ibuprofen and its derivatives has been proposed to lower the risk for developing Alzheimer's disease and clinical trials for this are performed.

Question 10: *How well are the inhibitors ranked by the scoring function?*
Question 11: *Would you say that the docking score of Ibuprofen is trustworthy if you compare it with the scores for the other compounds that are known inhibitors of acetylcholinesterase?*

Look at the hydrogen bonds between the ligands and the protein: Right-click on the group in the Workspace Navigator and click on View Poses. Ensure that Display H-bonds is chosen. Hydrogen bonds to the currently displayed pose are displayed as yellow dashed lines.
Count the hydrogen bonds between the ligands and the receptor by clicking Create property and selecting H-bonds and halogen bonds. An HBond property will be added to the Project Table.

Question 12: *How many hydrogen bonds do you see? Which atoms are involved? (You can choose which interactions are displayed in the Workspace through Pose Viewer.)*

Let's assess the fit of the ligand in the pocket by looking at steric clashes between ligand and protein. In Maestro it is possible to calculate the number of "good", "bad", and "ugly" ligand-protein contacts. [1] An ugly contact essentially means that there is severe overlap between the atoms.
If Display contacts was not selected in View Poses during the previous step, select it and click on Create Property again. There should be three new properties in the Project Table: Good, Bad, and Ugly. You can visualise them individually in the Workspace by ticking the appropriate squares in Pose Viewer.

[1] The criteria in Maestro for whether a contact is defined as good, bad or ugly is based on the formula C = D12 / (R1 + R2), where D12 is the distance between atomic centres 1 and 2, and R1 and R2 are the radii of the atomic centres. C has default values of 1.30, 0.89 and 0.75 for good, bad, and ugly contacts, respectively. Distances greater than C(good) are not marked and those less than C(good) but greater than c(bad) are marked as "good", etc.

Question 13: *How many good, bad, and ugly contacts do you see? Which types of residues are involved?*

It can also be useful to create receptor and ligand surfaces using the Surface (Binding Site) button.

To summarise today's session:
Question 14: *What can you generally say about the binding pattern between the ligands and the enzyme?*

Day 2. Design your own ligand

Design your own acetylcholinesterase inhibitors by using the Build and Fragments Panel or the 2D Sketcher under Edit. Let yourself be inspired by the known inhibitors, the binding cavity and the docking results and conclusion from the last lab session. Don't forget to minimise your ligands. Dock them to the grid you generated in the last lab session and compare the Glide scores. If a design fails, think about why and how you can improve the structure to get a better scoring. In the end, you should have at least 3 improved ligands based on different starting structures.

Present your results in a written report (see instructions on next page) and hand it in before the exam.

Written report

The written report should be about three pages (not including pictures) and handed in before the exam. It should include:

- **Introduction**

  Short introduction about the target protein; write about acetylcholine and why it is important, about the receptor, and how this is connected to Alzheimer's disease.

- **Method**

  Briefly describe the idea behind molecular docking and why it is useful and provide some background on Glide.

- **Results**

  Present your inhibitors (with structures shown) and how they bind to the protein according to Glide, including any "failed" structures. You should have at least 3 improved ligands. Present the scores in a table.

- **Discussion**

  Compare your inhibitor with already existing inhibitors. What was your rationale for the design? What did you intend to achieve? If your design didn't work as expected, how did you modify the structure to improve the score?

- **Conclusion**

  Short summary of results.

- **References**

  Include at least 3-5 scientific and relevant references in the format:

  [1] Richard A. Friesner et. al., Glide: A New Approach for Rapid, Accurate Docking and Scoring. 1. Method and Assessment of Docking Accuracy, Journal of Medicinal Chemistry 2004 47 (7), 1739-1749

- **Appendix**

  Provide the Questions and Answers from the Day 1 session.